26900366 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours) REGULAR EXAMINATION October 2026

Fifth SEMESTER

Skill Enhancement Course (SEC) - MG5SECMAT300 - INTRODUCTION TO PYTHON FOR MATHEMATICAL COMPUTATION

(2024 ADMISSION ONWARDS)

Duration: 1 hour 15 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions

Answer any **5** questions

Each question carries **2** marks

1. Write the output of the following code: [U] / [CO1]

$$mylist = range(5)$$
$$print\ mylist$$
$$for\ item\ in\ mylist:$$
$$print\ item$$

2. Explain an immutable type with an example. [U] / [CO2]

3. What is a method in Python? How is a method invoked on an object? [U] / [CO1]

4. Write the Python statement to compute the cross product of A = [4, 2, 1] and B = [3, 6, 5]. [A] / [CO3]

5. Which NumPy method is used to create an independent copy of an array? [A] / [CO3]

6. Write the Python statement to save the array $a = arange(10)$ into a file named `myfile.dat`. [U] / [CO3]

7. Write the Cartesian equation of an astroid. [K] / [CO4]

8. Define Mandelbrot set. [K] / [CO4]

[2x5 = 10]

Part B

Short Essay Type Questions

Answer any **4** questions

Each question carries **5** marks

9. Consider the following Python statements: [U] / [CO1]

```
x = 10
print(type(x))
x = 10.5
print(type(x))
x = "Python"
print(type(x))
```

Explain the change in the type of *x* during execution. What does this demonstrate about variables in Python?

10. Write a Python script to display the matrix $\begin{bmatrix} 2 & 4 \\ 6 & 8 \end{bmatrix}$ as the element-wise sum and element-wise product of any 2 matrices. [S] / [CO3]

11. Explain the function `arange()` with its syntax. Write a Python statement to generate the array `[2.0, 2.5, 3.0, 3.5, 4.0, 4.5]`. [A] / [CO3]

12. Write a Python program to find the square of every element in the array $A = [1, 2, 3, 4, 5]$ using a user-defined vectorized function. [A] / [CO3]

13. Write a Python program to plot a Square wave using Fourier series, number of terms shhould be a variable. [S] / [CO4]

14. Write a Python program to plot a 3D line plot using $x = \cos(\phi), y = \sin(\phi), z = \sin(4\phi)$ where $0 \leq \phi \leq 2\pi$, with the axes labelled. [U] / [CO4]

[5x4 = 20]

Part C

Essay Type Questions

Answer any **2** questions

Each question carries **10** marks

15. (a) Write a Python program to calculate the factorial of a number using recursion. [A] / [CO1]

(b) Write a Python program to print the numbers from 1 to 20 using a for loop. Use the `continue` statement to skip the multiples of 3.

16. Explain in detail the different methods to create arrays using the NumPy module, providing relevant examples. [U] / [CO3]

17. What is the difference between the dot product and the cross product? Write a Python program for the cross product $A \times B$ and dot product $A.B$ where $A = [8, 4, 6], \quad B = [10, 6, 8]$ and show the result. [E] / [CO3]

18. (a) Write a Python program using the matplotlib module to plot the functions $f(x) = \sin^2 x$ and $g(x) = \sin x^3$ on a single coordinate system. Both curves must be plotted over the interval $[-\pi, \pi]$ using 200 data points to ensure smoothness, with $g(x)$ explicitly colored red. [A] / [CO4]
(b) Write a Python program using the matplotlib module to plot the functions $y_1 = t^2$ and $y_2 = t^3$ on the same graph. The variable t should range from 0.0 up to (but excluding) 5.0 with a step size of 0.2. Use 'x' markers for y_1 and red circles for y_2.

[10x2 = 20]